

SIM1003 Calculus II
Tutorial 10

All even-numbered questions in the following exercises

Exercise 12.1:

In Exercises 1–4, find the given limits.

1. $\lim_{t \rightarrow \pi} \left[\left(\sin \frac{t}{2} \right) \mathbf{i} + \left(\cos \frac{2}{3} t \right) \mathbf{j} + \left(\tan \frac{5}{4} t \right) \mathbf{k} \right]$
2. $\lim_{t \rightarrow -1} \left[t^3 \mathbf{i} + \left(\sin \frac{\pi}{2} t \right) \mathbf{j} + (\ln(t + 2)) \mathbf{k} \right]$
3. $\lim_{t \rightarrow 1} \left[\left(\frac{t^2 - 1}{\ln t} \right) \mathbf{i} - \left(\frac{\sqrt{t} - 1}{1 - t} \right) \mathbf{j} + (\arctan t) \mathbf{k} \right]$
4. $\lim_{t \rightarrow 0} \left[\left(\frac{\sin t}{t} \right) \mathbf{i} - \left(\frac{\tan^2 t}{\sin 2t} \right) \mathbf{j} - \left(\frac{t^3 - 8}{t + 2} \right) \mathbf{k} \right]$

Motion in Plane

In Exercises 5–8, $\mathbf{r}(t)$ is the position of a particle in the xy -plane at time t . Find an equation in x and y whose graph is the path of the particle. Then find the particle's velocity and acceleration vectors at the given value of t .

5. $\mathbf{r}(t) = (t + 1)\mathbf{i} + (t^2 - 1)\mathbf{j}; t = 1$
6. $\mathbf{r}(t) = \frac{t}{t+1}\mathbf{i} + \frac{1}{t}\mathbf{j}; t = -\frac{1}{2}$
7. $\mathbf{r}(t) = e^t \mathbf{i} + \frac{2}{9}e^{2t}\mathbf{j}; t = \ln 3$
8. $\mathbf{r}(t) = (\cos 2t)\mathbf{i} + (3\sin 2t)\mathbf{j}; t = 0$

Motion in Space

In Exercises 13–18, $\mathbf{r}(t)$ is the position of a particle in space at time t . Find the particle's velocity and acceleration vectors. Then find the particle's speed and direction of motion at the given value of t . Write the particle's velocity at that time as the product of its speed and direction.

13. $\mathbf{r}(t) = (t + 1)\mathbf{i} + (t^2 - 1)\mathbf{j} + 2t\mathbf{k}; t = 1$
14. $\mathbf{r}(t) = (1 + t)\mathbf{i} + \frac{t^2}{\sqrt{2}}\mathbf{j} + \frac{t^3}{3}\mathbf{k}; t = 1$
15. $\mathbf{r}(t) = (2\cos t)\mathbf{i} + (3\sin t)\mathbf{j} + 4t\mathbf{k}; t = \pi/2$
16. $\mathbf{r}(t) = (\sec t)\mathbf{i} + (\tan t)\mathbf{j} + \frac{4}{3}t\mathbf{k}; t = \pi/6$
17. $\mathbf{r}(t) = (2\ln(t + 1))\mathbf{i} + t^2\mathbf{j} + \frac{t^2}{2}\mathbf{k}; t = 1$
18. $\mathbf{r}(t) = e^{-t}\mathbf{i} + (2\cos 3t)\mathbf{j} + (2\sin 3t)\mathbf{k}; t = 0$

In Exercises 19–22, $\mathbf{r}(t)$ is the position of a particle in space at time t . Find the angle between the velocity and acceleration vectors at time $t = 0$.

19. $\mathbf{r}(t) = (3t + 1)\mathbf{i} + \sqrt{3}t\mathbf{j} + t^2\mathbf{k}$

20. $\mathbf{r}(t) = \left(\frac{\sqrt{2}}{2}t\right)\mathbf{i} + \left(\frac{\sqrt{2}}{2}t - 16t^2\right)\mathbf{j}$

21. $\mathbf{r}(t) = (\ln(t^2 + 1))\mathbf{i} + (\arctan t)\mathbf{j} + \sqrt{t^2 + 1}\mathbf{k}$

22. $\mathbf{r}(t) = \frac{4}{9}(1 + t)^{3/2}\mathbf{i} + \frac{4}{9}(1 - t)^{3/2}\mathbf{j} + \frac{t}{3}\mathbf{k}$

Tangents to Curves

In Exercises 23–26, find parametric equations for the line that is tangent to the given curve $\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$ at the given parameter value $t = t_0$.

23. $\mathbf{r}(t) = (\sin t)\mathbf{i} + (t^2 - \cos t)\mathbf{j} + e^t\mathbf{k}; t_0 = 1$

24. $\mathbf{r}(t) = t^2\mathbf{i} + (2t - 1)\mathbf{j} + t^3\mathbf{k}; t_0 = 2$

25. $\mathbf{r}(t) = \ln t\mathbf{i} + \frac{t-1}{t+2}\mathbf{j} + t\ln t\mathbf{k}; t_0 = 1$

26. $\mathbf{r}(t) = (\cos t)\mathbf{i} + (\sin t)\mathbf{j} + (\sin 2t)\mathbf{k}; t_0 = \pi/2$

In Exercises 27–30, find the value(s) of t so that the tangent line to the given curve contains the given point.

27. $\mathbf{r}(t) = t^2\mathbf{i} + (1 + t)\mathbf{j} + (2t - 3)\mathbf{k}; (-8, 2, -1)$

28. $\mathbf{r}(t) = t\mathbf{i} + 3\mathbf{j} + \left(\frac{2}{3}t^{3/2}\right)\mathbf{k}; (0, 3, -8/3)$

29. $\mathbf{r}(t) = 2t\mathbf{i} + t^2\mathbf{j} - t^2\mathbf{k}; (0, -4, 4)$

30. $\mathbf{r}(t) = -t\mathbf{i} + t^2\mathbf{j} + (\ln t)\mathbf{k}; (2, -5, -3)$

Exercise 12.2:

Integrating Vector-Valued Functions

Evaluate the integrals in Exercises 1–10.

1. $\int_0^1 [t^3 \mathbf{i} + 7\mathbf{j} + (t + 1)\mathbf{k}] dt$

2. $\int_1^2 \left[(6 - 6t)\mathbf{i} + 3\sqrt{t}\mathbf{j} + \left(\frac{4}{t^2}\right)\mathbf{k} \right] dt$

3. $\int_{-\pi/4}^{\pi/4} [(\sin t)\mathbf{i} + (1 + \cos t)\mathbf{j} + (\sec^2 t)\mathbf{k}] dt$

4. $\int_0^{\pi/3} [(\sec t \tan t)\mathbf{i} + (\tan t)\mathbf{j} + (2 \sin t \cos t)\mathbf{k}] dt$

5. $\int_1^4 \left[\frac{1}{t}\mathbf{i} + \frac{1}{5-t}\mathbf{j} + \frac{1}{2t}\mathbf{k} \right] dt$

6. $\int_0^1 \left[\frac{2}{\sqrt{1-t^2}}\mathbf{i} + \frac{\sqrt{3}}{1+t^2}\mathbf{k} \right] dt$

7. $\int_0^1 [te^{t^2}\mathbf{i} + e^{-t}\mathbf{j} + \mathbf{k}] dt$

8. $\int_1^{\ln 3} [te^t\mathbf{i} + e^t\mathbf{j} + \ln t\mathbf{k}] dt$

9. $\int_0^{\pi/2} [\cos t\mathbf{i} - \sin 2t\mathbf{j} + \sin^2 t\mathbf{k}] dt$

10. $\int_0^{\pi/4} [\sec t\mathbf{i} + \tan^2 t\mathbf{j} - t \sin t\mathbf{k}] dt$

Initial Value Problems

Solve the initial value problems in Exercises 11–16 for $\mathbf{r}$ as a vector function of t .

11. Differential equation: $\frac{d\mathbf{r}}{dt} = -t\mathbf{i} - t\mathbf{j} - t\mathbf{k}$

Initial condition: $\mathbf{r}(0) = \mathbf{i} + 2\mathbf{j} + 3\mathbf{k}$

12. Differential equation: $\frac{d\mathbf{r}}{dt} = (180t)\mathbf{i} + (180t - 16t^2)\mathbf{j}$

Initial condition: $\mathbf{r}(0) = 100\mathbf{j}$

13. Differential equation: $\frac{d\mathbf{r}}{dt} = \frac{3}{2}(t + 1)^{1/2}\mathbf{i} + e^{-t}\mathbf{j} + \frac{1}{t + 1}\mathbf{k}$

Initial condition: $\mathbf{r}(0) = \mathbf{k}$

14. Differential equation: $\frac{d\mathbf{r}}{dt} = (t^3 + 4t)\mathbf{i} + t\mathbf{j} + 2t^2\mathbf{k}$

Initial condition: $\mathbf{r}(0) = \mathbf{i} + \mathbf{j}$

15. Differential equation:

$$\frac{d\mathbf{r}}{dt} = (\tan t)\mathbf{i} + \left(\cos\left(\frac{1}{2}t\right)\right)\mathbf{j} - (\sec 2t)\mathbf{k}$$

Initial condition: $\mathbf{r}(0) = 3\mathbf{i} - 2\mathbf{j} + \mathbf{k}$

16. Differential equation:

$$\frac{d\mathbf{r}}{dt} = \left(\frac{t}{t^2 + 2}\right)\mathbf{i} - \left(\frac{t^2 + 1}{t - 2}\right)\mathbf{j} + \left(\frac{t^2 + 4}{t^2 + 3}\right)\mathbf{k}$$

Initial condition: $\mathbf{r}(0) = \mathbf{i} - \mathbf{j} + \mathbf{k}$